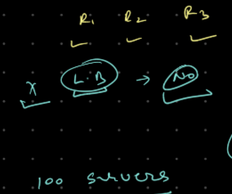
Consistent Hashing ⇒ sharding

Redis → Buckets → 1 Bucket per person

Redis instance / server → 200K

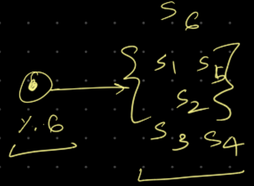
1B → 10% of 1B → 100M → Redis

R1 R2 R3 → horizontal scaling ?



→ sharding → 1→100K R1 101→200K R2





100 servers

userid ⇒ h(uid) ⇒ 1067 %. 100 → 0 → 99 ← 67

↳ 67th server

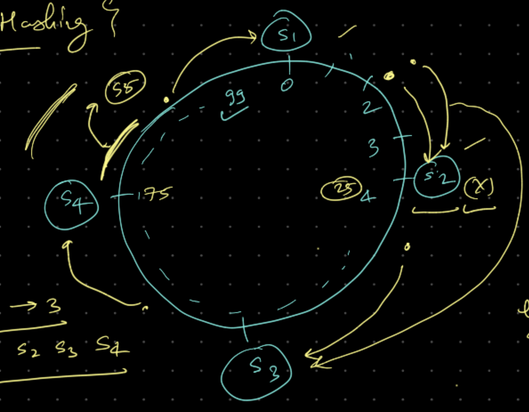
Problem



Removing / Adding ⇒ {Traffic Redistributes}

Goal → {minimize this redistribution}

{Consistent Hashing}



4 servers

100 slots

→ req → h(req)

h %. 100 → 0 → 99

h %. 100 → 0→25
0→25

Removing a server → load on a particular server spiked



S1 → 0 0→100 [0, 15, 75]

S2 → 25

S3 → 50

S4 → 75

J.w.T

↳ token + JSON + web App

① JSON · web Token → stateless ①

② J.w.T hide data ②



